Remote Direct Code Execution (HotNets '25)

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Presented by: Ian Chen

Yibo Huang et al. (Nov. 2025). "Remote Direct Code Execution". In: *Proceedings of the 24th ACM Workshop on Hot Topics in Networks*. HotNets '25. ACM, pp. 300–307

Runtime Programmable Networks

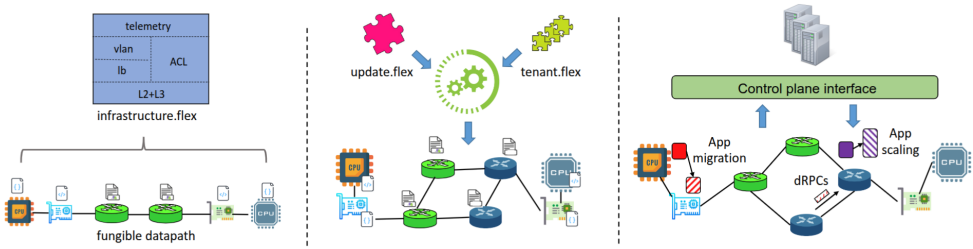


Figure: **FlexNet:** Figure 1 from (Xing et al. 2021). A central controller allows for *real-time management* of cross-layer configuration. This provides *disruption-free changes* of network devices.

Jiarong Xing et al. (Nov. 2021). "A Vision for Runtime Programmable Networks". In: *Proceedings of the Twentieth ACM Workshop on Hot Topics in Networks*. HotNets '21. ACM, pp. 91–98

Runtime Programmable Networks (eBPF)

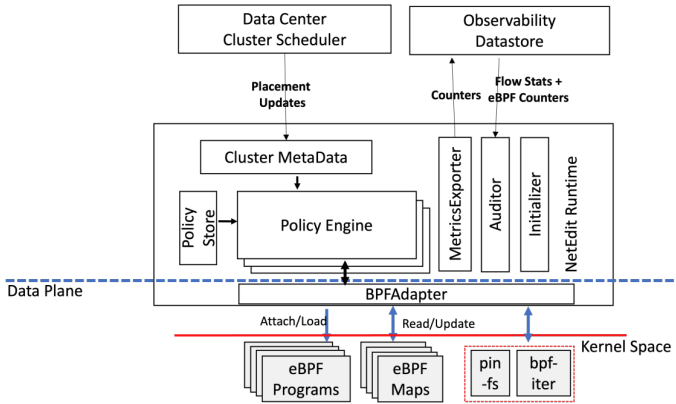


Figure: **NetEdit:** Figure 2 from (Benson et al. 2024). eBPF is used in NetEdit for observability, diverse priorities / network characteristics, and constant evolution.

Theophilus A. Benson et al. (Aug. 2024). "NetEdit: An Orchestration Platform for eBPF Network Functions at Scale". In: *Proceedings of the ACM SIGCOMM 2024 Conference*. ACM SIGCOMM '24. ACM, pp. 721–734

Runtime Programmable Networks (service meshes)

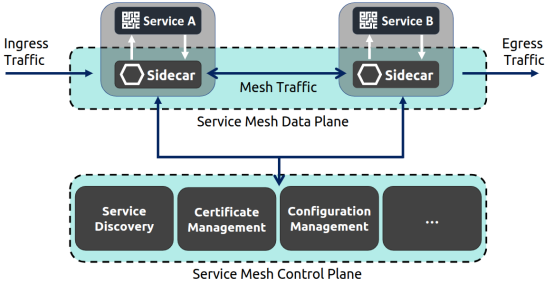


Figure: **Istio Service mesh**: Figure 1 from (Ashok, Godfrey, and Mittal 2021).

Sachin Ashok, P. Brighten Godfrey, and Radhika Mittal (Nov. 2021). "Leveraging Service Meshes as a New Network Layer". In: *Proceedings of the Twentieth ACM Workshop on Hot Topics in Networks*. HotNets '21. ACM, pp. 229–236

Istio (2026). *Extensibility*. https://istio.io/latest/docs/concepts/extensibility/

Runtime Programmable Networks (service meshes)

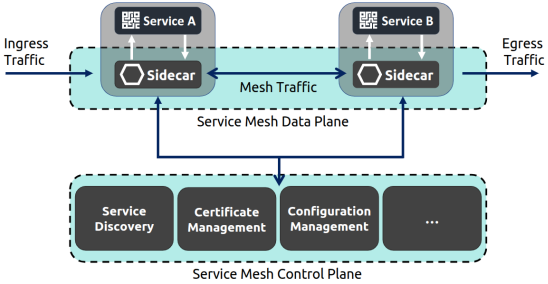


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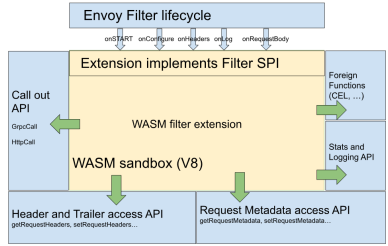
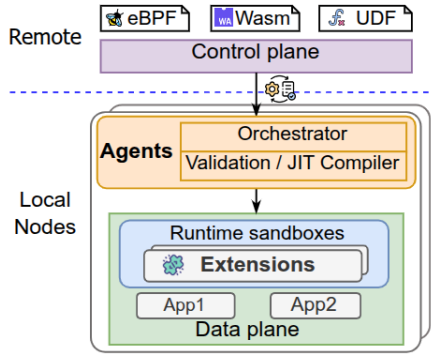


Figure: **Extensible Istio envoys**: Extensibility in sidecars using WASM filters (Istio 2026).

Sachin Ashok, P. Brighten Godfrey, and Radhika Mittal (Nov. 2021). "Leveraging Service Meshes as a New Network Layer". In: *Proceedings of the Twentieth ACM Workshop on Hot Topics in Networks*. HotNets '21. ACM, pp. 229–236

Istio (2026). *Extensibility*. https://istio.io/latest/docs/concepts/extensibility/

Limitations of Existing Solutions



(a) SOTA: Agent rollout

Figure: **SOTA agent rollout**: Extensions must be compiled and configured on the local node, typically with agents for managing hooks. Fig, 1 from (Huang et al. 2025).

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Limitations of Existing Solutions (cont.)

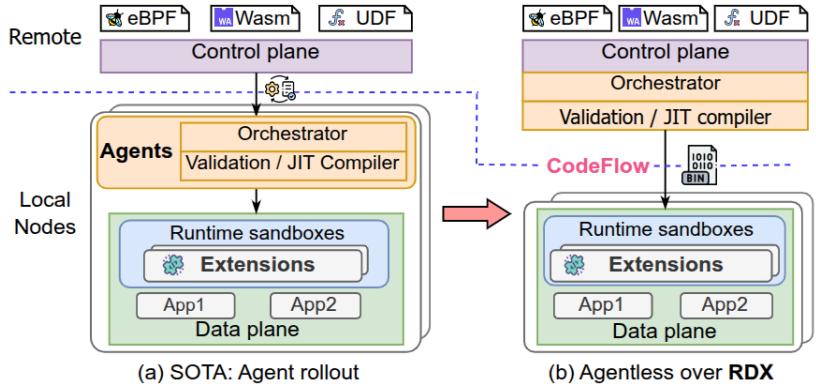
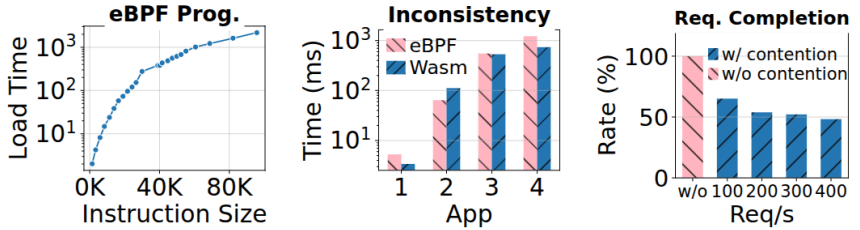


Figure: **Agentless over RDX**: Orchestrator, validation/JIT done on the remote node, deployed onto local through RDMA on management stub. Fig. 1 from (Huang et al. 2025).

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Limitations of Existing Solutions (cont.)



(a) eBPF program injection overhead **(b) Update inconsistency duration** **(c) Application perf. degradation**

Figure: **Overhead**: a) eBPF load time is $O(\text{ms})$ b) updating many services (4 apps, 33 microservices) introduces inconsistency c) updates cause 40-50% degradation in application performance. Fig. 2 from (Huang et al. 2025).

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RDX

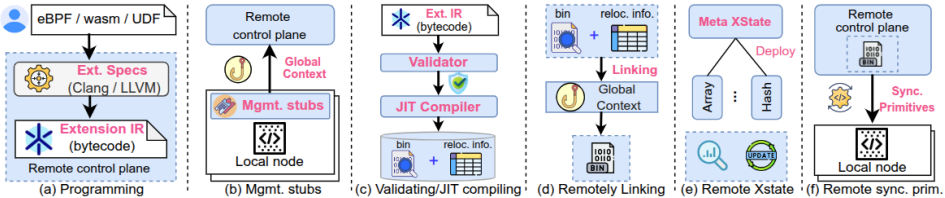


Figure: **RDX solution sketch**: Fig. 3 from (Huang et al. 2025).

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RDX

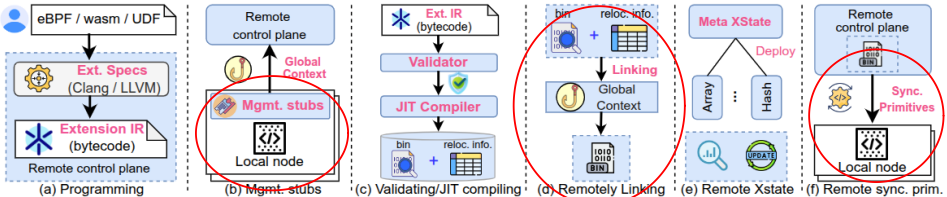


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RDX Results

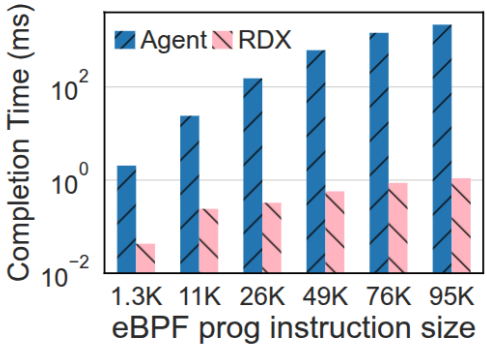


Figure: **RDX reduces local eBPF overhead**: Fig. 4 from (Huang et al. 2025).

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My thoughts

- Motivation for implementing RDX via RDMA (over say, RPCs) is not clear
  - authors say want $O(\mu s)$ auto-scaling and roll-back
- A lot of emphasis is put on the *agentless architecture*
  - why agents in the first place?
- Security concerns are not addressed (*remote code execution*!)